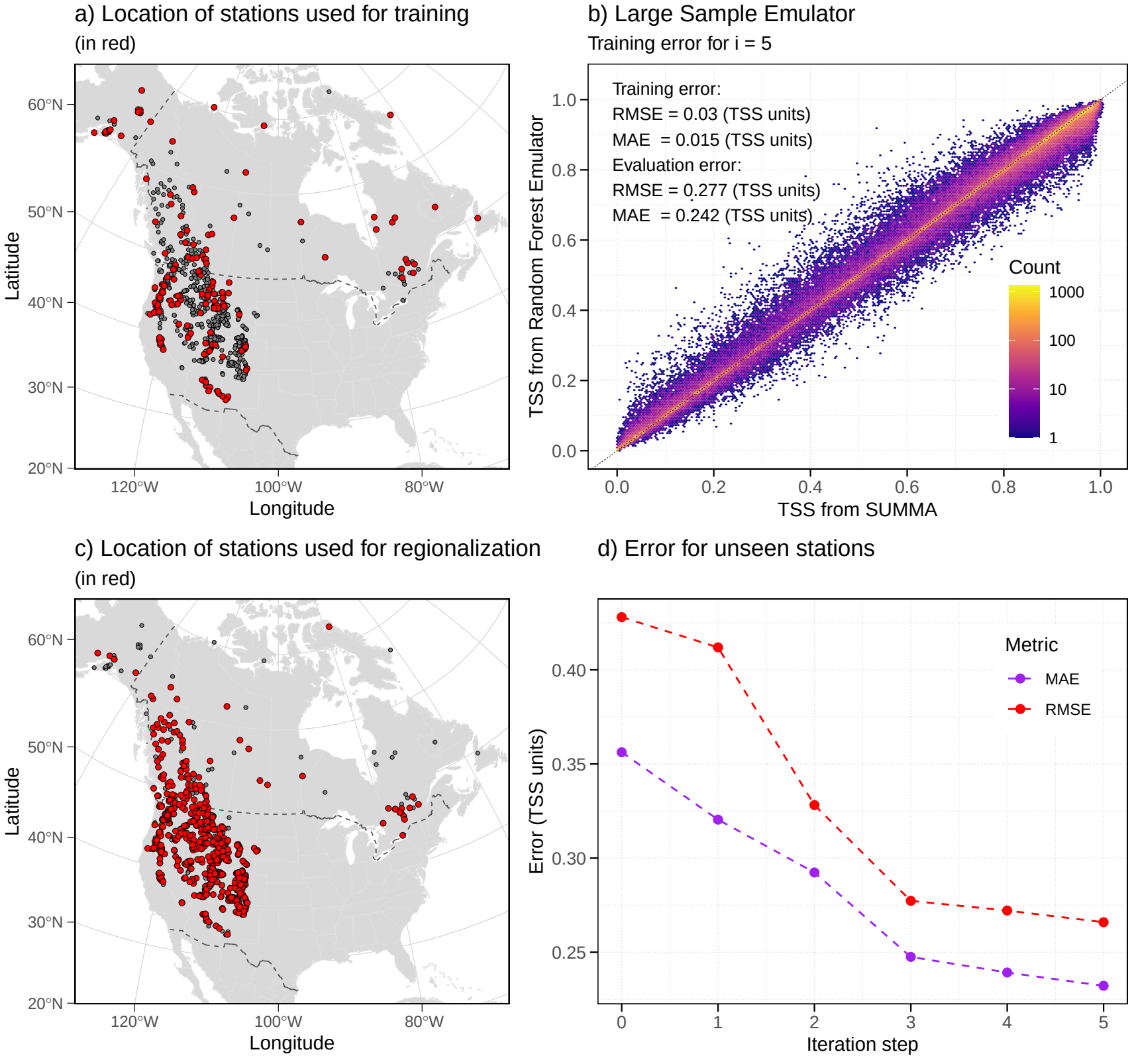

a) Location of stations used for training
(in red)
60°N
50°N
40°N
30°N
20°N
Latitude
120°W
100°W
80°W
Longitude
b) Large Sample Emulator
Training error for i = 5
Training error:
RMSE = 0.03 (TSS units)
MAE = 0.015 (TSS units)
Evaluation error:
RMSE = 0.277 (TSS units)
MAE = 0.242 (TSS units)
Count
1000
100
10
1
TSS from Random Forest Emulator
TSS from SUMMA
0.0
0.2
0.4
0.6
0.8
1.0
c) Location of stations used for regionalization
(in red)
d) Error for unseen stations
Metric
MAE
RMSE
Error (TSS units)
0.40
0.35
0.30
0.25
Iteration step
0
1
2
3
4
5